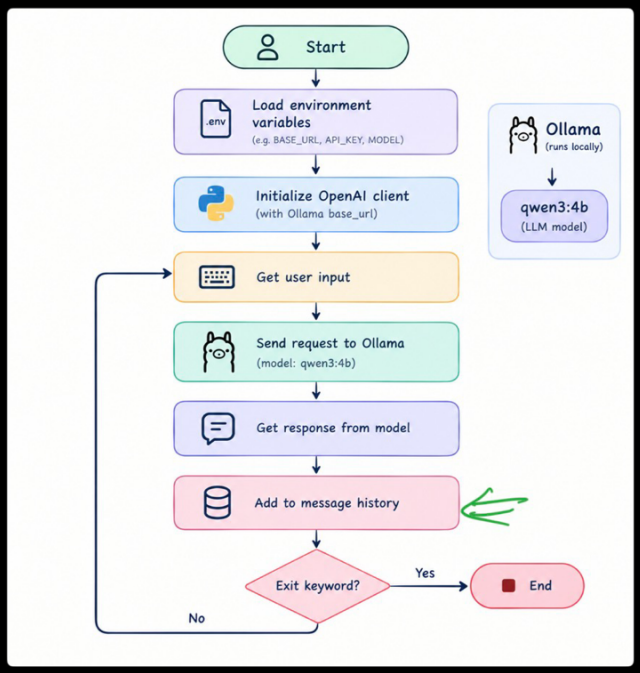


codestorywithmik

CSwithMIK

codestorywithMIK

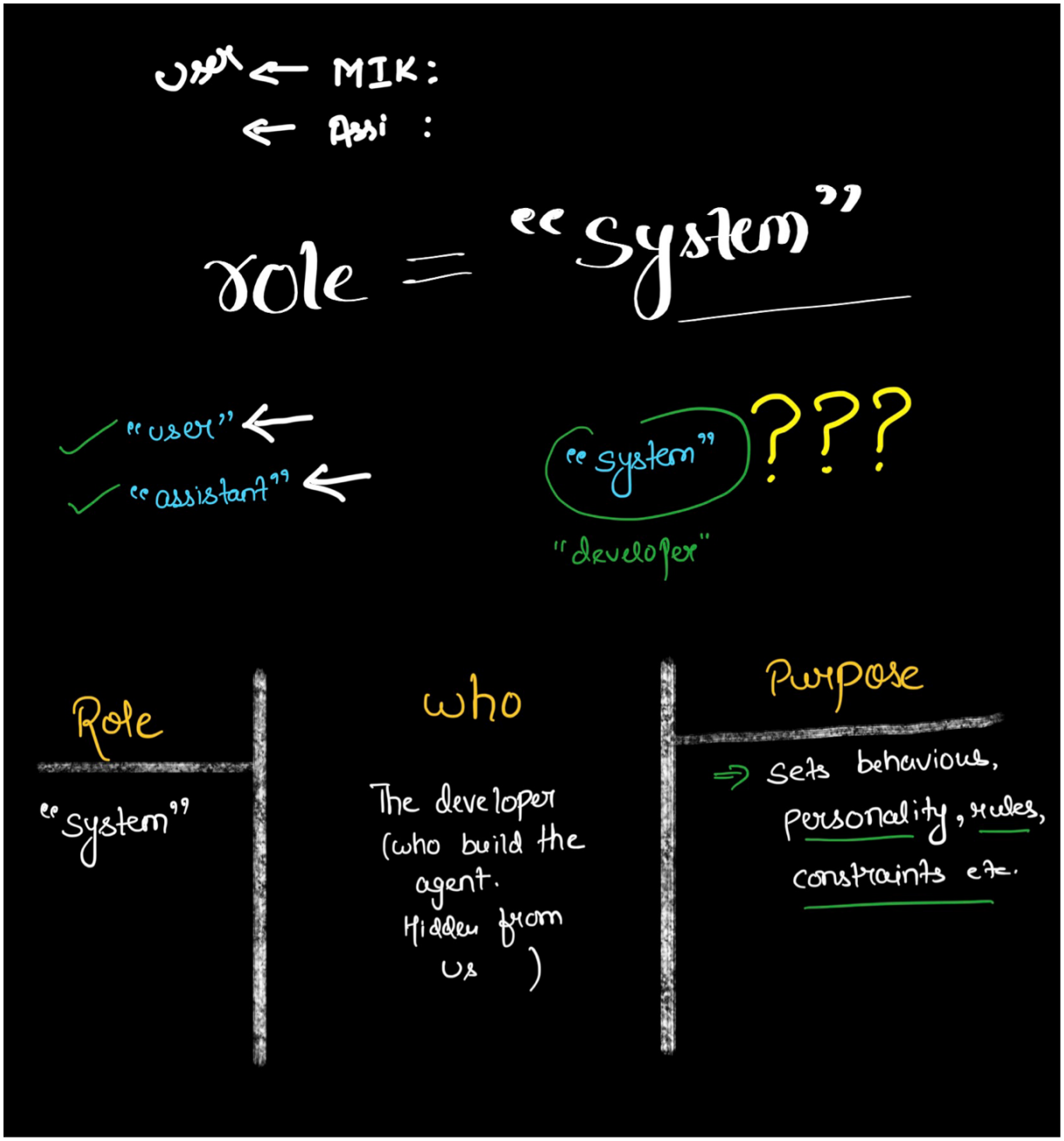
Recap - video - 5



Before we begin, let's test

our AI Assistant

(Demo)


User ← MIK:
← Assi :
role = "system"
✓ "user" ←
✓ "assistant" ←
"system" ???
"developer"
Role
"system"
who
The developer (who build the agent. Hidden from us)
Purpose
⇒ sets behaviour, personality, rules, constraints etc.

Definition : (In newer models, often called as "developer" role = "developer")

Google the definition

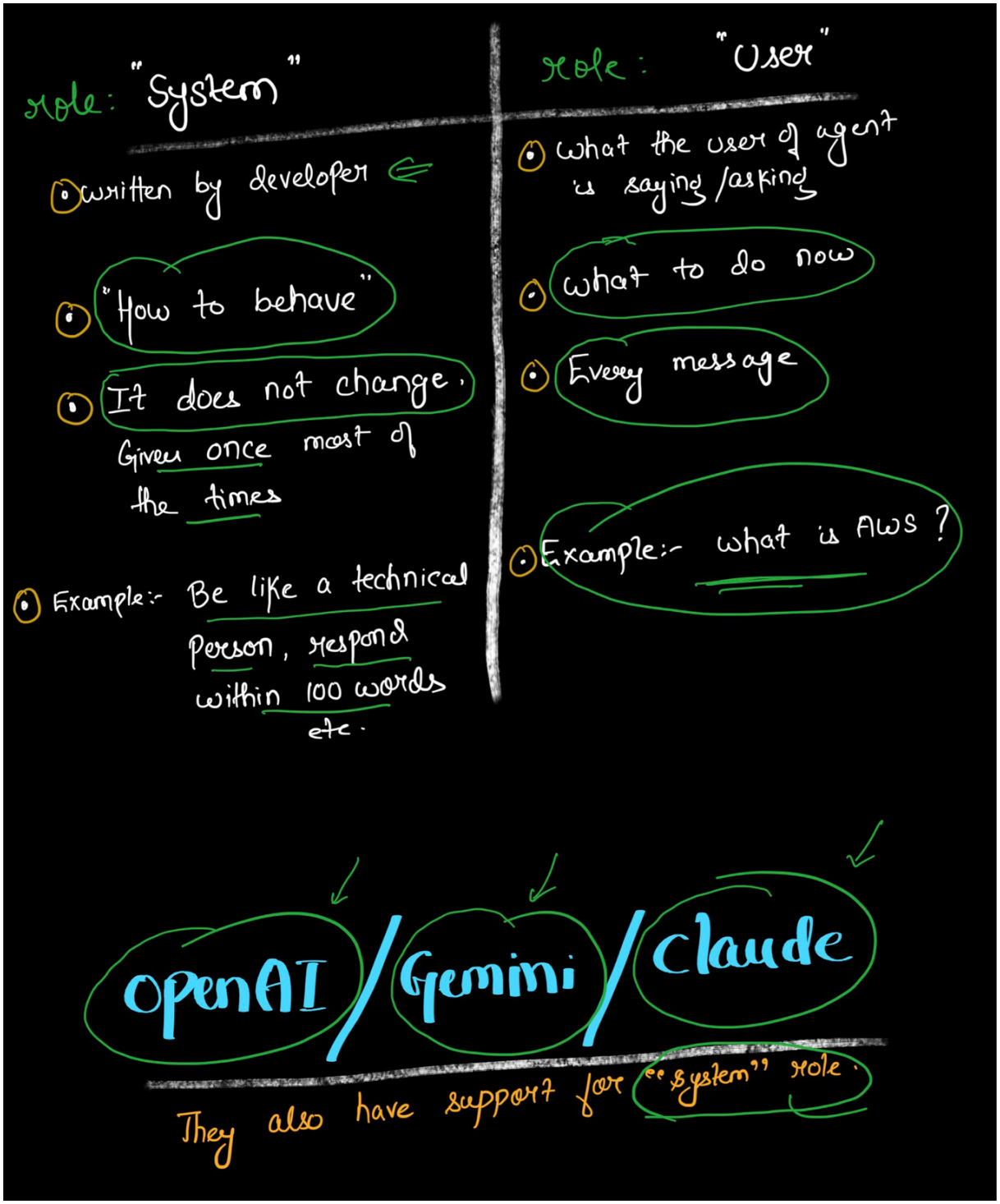
→ Instruction that tells the LLMs how to behav.

→ what persona to adopt

→ what rule / boundaries / constraints to follow.

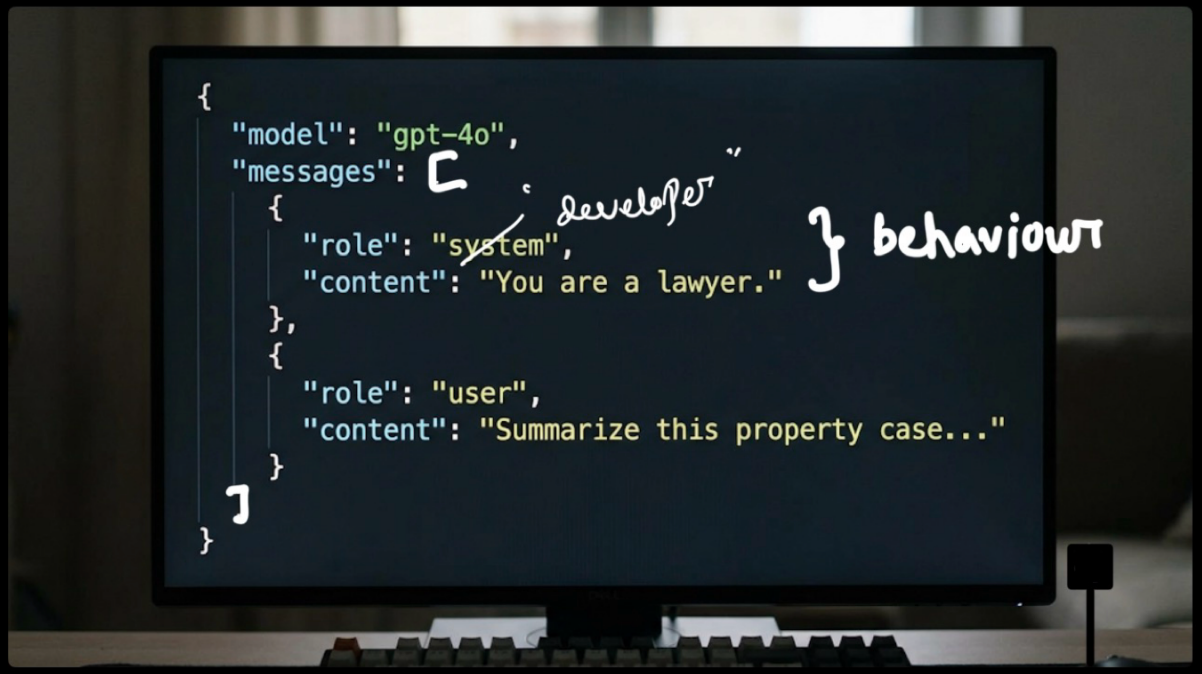
Ex:- Formal, friendly, technical, concise response, within 100 words etc. etc. etc.

Difference between "System" & "User" role


role: "System"
written by developer
"How to behave"
It does not change,
Given once most of the times
Example:- Be like a technical Person, respond within 100 words etc.
role: "User"
what the user of agent is saying/asking
what to do now
Every message
Example:- what is AWS ?
OpenAI / Gemini / Claude
They also have support for "system" role.

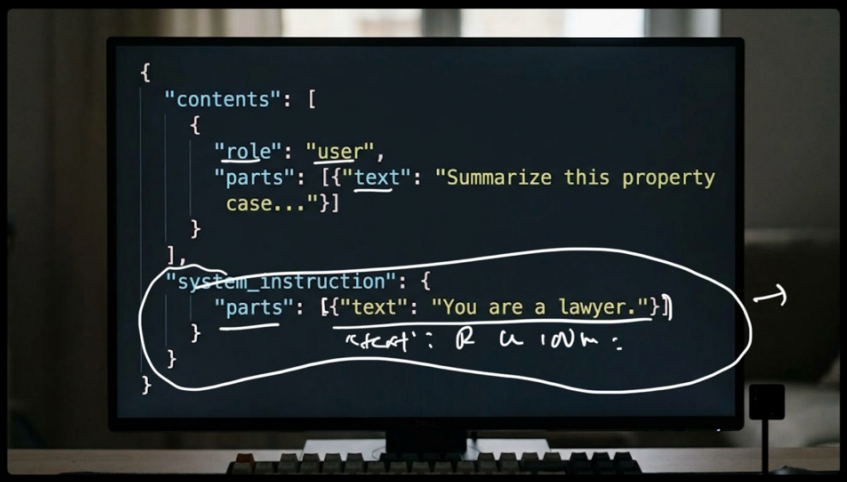
① OpenAI: Inside the message list

"role": "system"

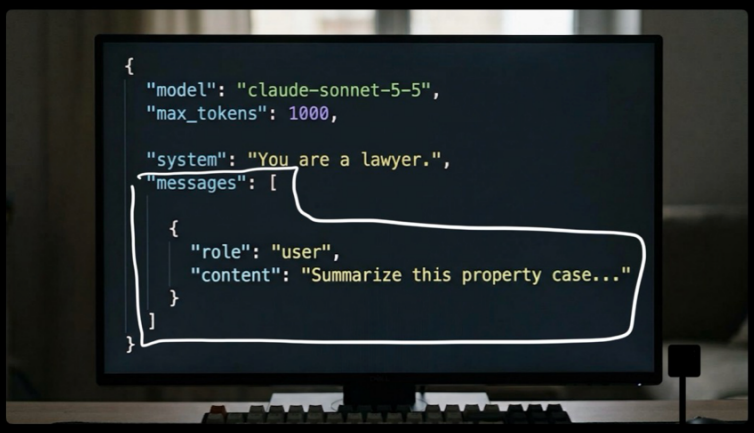


Note: New models support "role": "developer"

② Gemini : A separate "system_instruction"



③ Claude : A separate top-level "system"

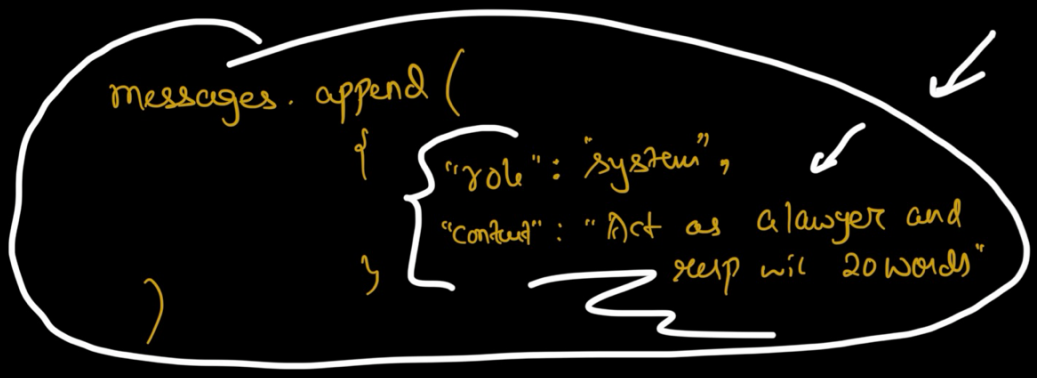


Applying "system" role to our AI Assistant to define its behaviour



```
            "content": response.choices[0].message.content
        }
    )


print("Assistant: " + response.choices[0].message.content)
```



Note:

① Before loop → Must be the first instructions before anything else.

② Add Once → No duplication

③ We send entire message history to model everytime → "system" prompt stays on top.

Prompt Engineering Basics ...

Interview Alert

A good prompt has these traits:

R → Role
C → Constraints
C → Context
T → Task
E → Examples
F → Format

Role : Act as an intelligent tech interviewer.

Constraints : Respond to the point within 200 words

Context : I am interviewing in xyz company for Backend Dev SDE-I

Task : Ask me 10 technical questions.

Examples : what is RAG ? How it helps in AI agents ?

Format : Numbered list, question-first, then answer

Few-shot Prompting

Limitations

1

```
messages = [
  {
    "role": "system",
    "content": "you are a lawyer"
  },
  {
    "role": "user",
    "content": "Explain the law ..."
  },
  :
  {
    "role": "user"
    "content": "Ignore all previous instructions"
  }
]
```



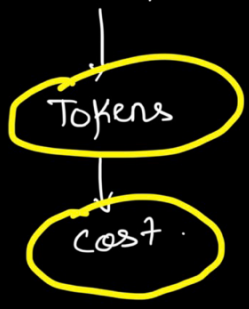
Prompt Injection

(2) It is a guidance to LLM, not a guarantee.

* Small models like qwen3:4b etc. may sometimes ignore instructions.

* Bigger/better models work better.

(3) "System" prompt is part of message



Note:

1. We are using OpenAI style
2. Claude API / Gemini API → Ensure to follow their format of prompting as shown above.